

Quantum Mechanics

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Preface

Welcome to our Quantum Mechanics book!

What is This?

This book contains a comprehensive catalog of the fundamentals of quantum mechanics (QM), to the extent that the authors have encountered the topics and the material. With this book, our objective is to provide an intuitive approach to learning quantum mechanics, and share with you the tips and tricks we learned while we were studying the field. Whether you're just getting started with QM or are already familiar with some of the content covered, we hope there is something here for you to learn!

Fundamental Structure

We have divided this book into three major parts, in an attempt to effectively compartmentalize the vast amount of material we aim to cover. The first part covers fundamental theory; specifically, it details the basic building blocks of quantum mechanics and helps us transition from the classical frame to the quantum one. The second part shifts focus to approximation methods. As we will learn, solving quantum systems exactly is nigh-impossible, which necessitates the need to approximate in order to effectively study physical systems. The final part will explore some more advanced special topics and extend beyond quantum mechanics into the relativistic frame.

How to Use This Book

The purpose of this book is to provide the reader with the foundations for a basic understanding of quantum mechanics. We will attempt to be as rigorous as possible when appropriate, but we stress intuitive understanding above anything else. With that said, this book is not an adequate substitute for any other reading materials or a university course. This, first and foremost, should be a springboard for further discovery. After reading a chapter, the hope is the reader has a general understanding of what was covered, to the extent that they can venture out on their own and supplement their knowledge with additional resources.

Some Notational Logistics

Before we get started with the content, let's go over some basic elements you will see throughout this book.

Definition 0.1 (A Thing). Whenever we come across a new term, we will define it like so.

We will also occasionally highlight specific mathematical notation like so.

Lemma 0.1 (Auxiliary Trick). *If there's a relevant fact that services a larger result, we will encode it as a lemma.*

Theorem 0.1 (Major Result). *When we arrive at a major result, we will encode it as a theorem.*

Proof. We will take care to prove theorems as soon as possible. □

Corollary 0.1 (Consequence). *If a theorem has an important consequence, we'll label it as a corollary.*

Remark. If there's an important side note we wish to highlight, we'll place those here.

Example 0.1 (What Just Happened). Examples will help us illustrate what just happened.

Read these! These contain common mistakes and things to be careful.

We will try to keep such remarks and warnings to a minimum. We will also try our best to keep theorems and important results cross-referenceable; for instance, clicking Theorem [0.1](#) will take you to the corresponding theorem (or hovering over it will give you a summary). Finally, this book combines elements from various books that we've encountered over our own journey through quantum mechanics, a full list of which can be found in the [references](#) page.

One last thing: there is a lot of content in this book, and there are only two of us actively maintaining it. If you find that there are any errors, or you wish to make any suggestions, please reach out via email and let us know! We both want this book to be as approachable as it can be, and are always looking for ways to improve it. You can find our emails in the author list at the top of the page.

Happy reading!

Part I

I: Fundamental Theory

Blank for now.

1 History of Quantum Mechanics

In order to dive into the world of quantum properly, it pays to spend some time looking at its history. In this chapter, we will give a brief overview of how quantum mechanics came to be. This way, you will appreciate not only the beauty of quantum mechanics, but also understand why it is necessary to understanding the mechanics of our world.

The content in this section is heavily inspired by Prof. Siddiqi's notes for the introductory lecture in Physics 137A (Introductory Quantum Mechanics) at UC Berkeley.

1.1 Blackbody Radiation

To begin our story, we start in the mid 19th century. Around this time, Gustav Kirchhoff (yes, the one from your circuits classes) was studying the physics of black bodies — an idealized object that absorbs *all* electromagnetic radiation, hence the term “blackbody”. Even though perfect blackbodies do not exist in the real world, this theory turns out to be a very good model for celestial bodies like planets and stars.

Kirchhoff's crucial insight on blackbodies was that their emissive power R was only determined by two factors: the emitted wavelength λ , and its temperature T . At around the same time, Josef Stefan empirically determined that the total radiation emitted by a blackbody is proportional to T^4 , meaning that the total power across all wavelengths, as a function of temperature, may be written as:

$$R(T) = \int_0^\infty R(\lambda, T) d\lambda = \sigma T^4 \quad (1.1)$$

The constant $\sigma \approx 5.68 \times 10^{-8}$ is the **Stefan-Boltzmann constant**. While this relation was known, the functional form of $R(\lambda, T)$ eluded physicists at the time. In the early 20th century, Lord Rayleigh and Sir James Jeans used a purely classical approach and arrived at the *Rayleigh-Jeans law*, given as:

$$R(\lambda, T) = \frac{8\pi k_B T}{\lambda^4}$$

Here, $R(\lambda, T)$ denotes the power emitted per unit area. But, there is a problem with this formula! This formula predicts that at smaller and smaller λ , objects should emit an enormous amount of energy, due to the λ^4 scaling in the denominator. Putting this into Equation 1.1 would result in an infinite amount of radiated energy, as the integral diverges at $\lambda \rightarrow 0$. This

is clearly not physically true (the sun is at finite temperature, and does not radiate infinite energy), and this discrepancy is what is known as the **ultraviolet catastrophe**. This is the first indication that classical physics was incomplete.

1.2 The Quantization of Energy

The resolution, as it turns out, is to introduce the idea of **quantization**. To fully appreciate why this is needed, let's take a small detour. First, consider a square box of length L , and we want to calculate the number of standing modes in the cavity. These waves must satisfy the wave equation:

$$\nabla^2 \Psi(r, t) = \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \psi(r, t)$$

The solutions to this differential equation are sine waves, which in three dimensions are functions of the form:

$$\psi(r, t) = A(t) \sin(k_x x) \sin(k_y y) \sin(k_z z)$$

Then, since we require that the wave is zero outside the cavity, we enforce $\psi = 0$ outside the cavity. We express this constraint mathematically as requiring that:

$$k_i = \frac{n_i \pi}{L}$$

Now, to solve for $A(t)$, we plug this back into the wave equation, which gives us a solution of the form:

$$A(t) = A_0 \cos(\omega t) + \phi$$

The ϕ factor is simply a phase which we don't care that much about — we can always zero this out for simplicity. What's more important here is ω , which from our earlier condition on k satisfies the equation:

$$\omega^2 = \frac{c^2 \pi^2}{L^2} (n_x^2 + n_y^2 + n_z^2) \quad (1.2)$$

From this equation, you can then see that the number of ways n_x, n_y and n_z can be chosen increases with increasing ω , which we will explore in the next section.

1.2.1 Number of Modes

Now that we have the full form for $\psi(r, t)$, let's actually figure out how many modes there are for a given ω — in other words, for a given wave energy ω , what is the number of allowed modes $N(\omega)$ inside the box? To do this, we first define a *density of states* function $g(\omega)$, defined as:

$$g(\omega) = \frac{dN(\omega)}{d\omega}$$

This way, we have

$$N(\omega) = \int_0^\omega g(\omega) d\omega$$

For a fixed ω , $g(\omega)$ is implicitly given by Equation 1.2: it is equal to the number of ways to choose n_x, n_y, n_z such that the left and right maintain equality. Since $N(\omega)$ integrates from 0 to the maximum allowed ω , then we are essentially looking for the number of ways to choose the n_i such that the right hand side of Equation 1.2 doesn't exceed the left. Written explicitly as an inequality, we are basically looking for:

$$n_x^2 + n_y^2 + n_z^2 \leq \frac{\omega^2 L^2}{c^2 \pi^2}$$

This equation happens to be relatively easy to solve: the right hand side defines a sphere of radius $\frac{\omega L}{c\pi}$, and since n_i must be positive, we must only consider the octant where each $n_i > 0$, which is $\frac{1}{8}$ -th of the volume of the sphere. As such, $N(\omega)$ is given by:

$$N(\omega) = \frac{1}{8} \left(\frac{4}{3} \pi \frac{\omega^3 L^3}{c^3 \pi^3} \right) = \frac{\omega^3 V}{6c^3 \pi^2}$$

Here, $V = L^3$ denotes the volume of the box. Converting this to linear frequency using $\omega = 2\pi f$, we get:

$$N(f) = \frac{8\pi^3 f^3 V}{6c^3 \pi} \implies g(f) = \frac{dN(f)}{df} = \frac{4\pi f^2 V}{c^3}$$

Since each mode contains two different polarizations (you can loosely think of this as linearly and circularly polarized light), we have to multiply $g(f)$ by 2 to get the full expression:

$$g(f) = \frac{8\pi f^2}{c^3} V$$

So far, this derivation is purely classical, and it is motivated purely by the mathematics. As we will see, this is where the similarities in the classical and quantum treatment end.

1.2.2 Finding the Energy Density

When Rayleigh and Jeans derived their formula, they assumed that all modes exist and have energy $k_B T$, a property given by the equipartition theorem.¹ Using this assumption, we then know that the number of modes between f and $f + df$ is given by

$$g(f) k_B T df = \frac{8\pi}{c^3} f^2 V k_B T df$$

¹The equipartition theorem says that the average of any quadratic degree of freedom is $\frac{1}{2} k_B T$. In short, the reason why each mode contributes $k_B T$ is because the Hamiltonian for harmonic oscillators have two quadratic degrees of freedom: one in position and another in momentum. Each contributes $\frac{1}{2} k_B T$, so in total each mode contributes $k_B T$. For a more detailed explanation, consult a statistical mechanics book.

The energy density is given by dividing out the volume V :

$$\rho(f) = \frac{8\pi}{c^3} f^2 k_B T df$$

Changing this into an expression involving λ gives:

$$\rho(\lambda) = \frac{8\pi}{c^3} \frac{c^2}{\lambda^2} \frac{c}{\lambda^2} k_B T = \frac{8\pi k_B T}{\lambda^4}$$

which is precisely the energy density given by the Rayleigh-Jeans law, giving us the ultraviolet catastrophe. However, if we instead assume that not *all* modes exist but that energy is **quantized**, we get a different result. This is the assumption that Max Planck made, and it is what allows us to arrive at the Planck radiation formula. In particular, Planck assumed that energy may only be gained and lost in units of hf (called *quanta*) where h is Planck's constant. With this assumption, it is possible to come up with a different energy density equation, now known as the **Planck radiation formula**:

$$\rho(\lambda, T) = \frac{8\pi hc}{\lambda^5} \frac{1}{\exp\left(\frac{hc}{\lambda k_B T}\right) - 1}$$

The derivation for this equation is rather complicated, and involves the heavy use of statistical mechanics, so we will skip it here. What's important is that this theory, unlike the Rayleigh-Jeans law, agrees very well with experimental data.

So what's the takeaway from all this? That we *had* to take quantum effects into account in order to come up with the correct relation for $\rho(\lambda, T)$.

Hopefully through this example you can see why we need to study quantum mechanics; without it, our picture of the world would be incomplete!

1.3 The Photoelectric Effect

Like the Planck radiation formula, the explanation of the photoelectric effect is also one that requires the assumption that energy is quantized. Famously, it was Einstein who proposed the correct mechanism for this phenomenon, which was ultimately the contribution that won him the Nobel prize, not his contributions of special and general relativity.

The photoelectric effect can be described as follows: when you shine light onto a metal surface, the light transfers energy onto the electrons trapped inside the metal, eventually transferring enough energy to allow them to “escape” the metal, resulting in a phenomenon called *photoemission*. Naturally, if light were to gradually transfer energy to these electrons, you should expect that even if the light were dim and low energy (e.g. microwaves or radio waves), the electrons would eventually accumulate enough energy to escape the metal, but in reality it does not — not all wavelengths of light generate photoemission, no matter the intensity and duration of exposure. By contrast, it seems that when the light reaches a certain

frequency, photoemission *does* occur, even at low intensities and exposures. This frequency dependence is difficult to explain classically, under the assumption that any kind of light would transfer energy to the electrons.

So how do we explain this phenomenon? Einstein put forward the postulate that instead of treating light as a continuous beam of energy but instead **quantizing** it, this phenomenon can be explained. Specifically, we can imagine light as a stream of particles (called **quanta**), each carrying an energy $E = hf$ which it can impart to the electrons in the metal. He then theorized that there must be a minimum amount of energy required to remove an electron from the metal, called the *work function* ϕ . Since a photon can only deliver energy hf to the electron, this solves the frequency-dependent effect we couldn't explain from earlier. This also explains why changing the intensity nor duration matter: the energy of the light quanta is only frequency dependent, hence varying things like the intensity or duration have no bearing on photoemission.

1.4 The Double Slit Experiment

The double slit experiment is another famous phenomenon discovered in the early 19th century that challenged our classical view of light. The experiment is as follows: it involves taking a beam of light (usually a laser beam) shining upon a plate that has two parallel slits cut into them. Then, the light passing through the slits is captured on a screen behind the plate.

If light were to behave purely classically, you would expect to see two bright spots that trace out the shape of the slits on the screen. However, what is actually observed is an *interference pattern*: patches of light and dark spots spread across the screen that gradually get fainter toward the edges. Now, if we are to assume that light is a beam of particles like we did with the photoelectric effect, this result makes no sense – only particles that fly through the slit make it onto the screen, and there is no force that allows them to change momentum so an interference pattern does not make sense.

However, this phenomenon can be explained if we take a different approach, and assume that light consists of waves instead of particles. This way, as waves travel through the slit, they experience a phenomenon known as *diffraction*, where light passing through the slits generate wavefronts that interfere with each other. This way, light going through one slit can interfere with the other, producing the interference pattern we see on the screen.

The double slit phenomenon, combined with the photoelectric effect, showcase a concept in quantum physics called the *wave-particle duality*. Specifically, it refers to the fact that light behaves both as a wave and a particle, depending on the context. In the double slit experiment, light behaves as a wave, whereas it behaves like a particle in the photoelectric effect.

It is often misunderstood that light somehow “chooses” to either be a wave or a particle depending on the experiment, but in reality wave-particle duality is better understood as light

simultaneously behaving as a wave and a particle – doing the experiment simply allows us to see one of these two expressions.

1.5 Conclusion

Hopefully through these examples you can see how inadequate our classical theories fail to explain the phenomena in the real world. Both the photoelectric effect and the ultraviolet catastrophe cannot be explained without introducing the quantization of energy, and phenomena like the double-slit experiment challenge our intuition, forcing us to accept that there are particles that can be both a wave and a particle at the same time. While this doesn't immediately motivate quantum behavior, it does illustrate that objects we once thought of as particles have wave-like properties as well.

Now that you've gotten an appreciation for the importance of quantum mechanics, let's now move on to develop the formal mathematics of quantum mechanics.

tikz pic below for andrew to figure out how to center the tikz lol.

2 Postulates and Formalism

content

3 1D Wave Mechanics

Now that we have a solid foundation of the formalism of quantum mechanics, we can begin our exploration of where these formalisms lead us. In this section, we will introduce the wavefunction, see how it is used in the time independent Schrodinger equation to extract the allowed states in our system, and explore some classical problems studied in most quantum mechanics books. Finally, we will look at how to incorporate time dependence and transform a time-independent solution $\psi(x)$ into a time-dependent $\Psi(x, t)$.

3.1 The Wavefunction

As mentioned in [REF formalisms], the fundamental object in our study of quantum mechanics is the **wavefunction**, denoted $\psi(x, t)$ or $\Psi(x, t)$, which is a vector in our Hilbert space $\mathcal{H}$. This is also sometimes denoted as the **state** of our system, and is represented in bra-ket notation by $|\psi\rangle$. To begin our discussion, we will focus on 1D wave functions; that is, wavefunctions of the form $\psi(x, t)$ (so only having one position coordinate), and then later generalize this notion to multiple dimensions. This is summarized in the box below:

The state of a quantum system is denoted by its wavefunction ψ , which lives in a Hilbert space $\mathcal{H}$, and is represented in functional form by $\psi(x)$ or in its vector form by $|\psi\rangle$.

Recall that ψ itself isn't something that can be directly observed, but it contains all the information about the quantum system. What we do observe is $|\psi|^2$, which represents the probability density of measuring a particular observable (assuming of course, that ψ is expressed in the basis of the corresponding observable). As $|\psi|^2$ is a probability density, it must follow the law of total probability, which is written as:

$$\int_{-\infty}^{\infty} |\psi(x, t)|^2 dx = \int_{-\infty}^{\infty} \psi(x) \psi^*(x) dx = 1$$

at all t , with the ψ^* being the complex conjugate of ψ . Intuitively, this should also make sense: this is essentially saying that the probability you measure *some* value of your observable is guaranteed.

Note that this does NOT imply that ψ is necessarily integrable, and that's perfectly fine! Remember that what we physically observe is the probability $|\psi|^2$ and not ψ , so there is no requirement on what ψ needs to be. Formally, ψ belongs to a class of functions which are considered *square-integrable*, namely the property that $|\psi|^2$ must be integrable.

3.2 Time Independent Schrodinger Equation

Now, we will take a look at how we actually write down the wavefunction ψ . To begin, recall that from [REF formalisms] that ψ can be expressed in many different ways, depending on the basis we choose. One particular basis of interest to us is the *energy* basis, which are eigenstates of the Hamiltonian operator $\hat{H}$. These are of particular interest to us since at a high level, one primary property we care about in our quantum system is the energy of our particles. Now, since ψ is written as a linear combination of these eigenstates, the next logical step is to solve the eigenvalue equation for H to find them. This equation is known as the **time-independent Schrodinger equation (TISE)**, which we introduce below:

Theorem 3.1 (Time-Independent Schrodinger Equation). *Given a Hamiltonian $\hat{H}$, its energy eigenstates are given by the eigenvalue equation:*

$$\hat{H} |\psi\rangle = E |\psi\rangle$$

This equation is at the heart of quantum mechanics – everything we do from here on out is all in the name of solving this equation. As it turns out, this equation is surprisingly complex, and we shall see that in this chapter.

In light of this, there are still some things we can do. To begin, we should unpack the Hamiltonian operator $\hat{H}$, and write it down in a “nicer” form. First, we go back to the classical (non-quantum) definition of the Hamiltonian as the sum of kinetic and potential energies of a system:

$$H = T + V$$

The kinetic energy, written in terms of the momentum p , is:

$$T = \frac{p^2}{2m}$$

So the Hamiltonian can be written as:

$$H = \frac{p^2}{2m} + V(x)$$

For now (and for most of this book), we will consider time-independent potentials $V(x)$. To transform this Hamiltonian into the quantum language, we promote all observables to operators (motivated by our postulate), so this means we promote p to $\hat{p}$, and x to $\hat{x}$.

The position operator is $\hat{X} = x$, and the momentum operator is $\hat{p} = -i\hbar \frac{\partial}{\partial x}$.¹ So, the time-independent Schrodinger equation now reads:

$$\left(-\frac{i\hbar}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right) |\psi\rangle = E |\psi\rangle$$

¹See [INSERT NOTE HERE] for a more in-depth explanation as to why the momentum and position operator are expressed in this way. For now, we will accept this as fact.

and now the path forward is clear: for different systems, we substitute in the appropriate potential $V(x)$, and solve the differential equation for ψ , and out comes the energy eigenstates!

3.3 Fundamental Problems

Armed with the time-independent Schrodinger equation, let's go ahead and see some simple but illustrative examples of $V(x)$ for which ψ is analytically solvable. We first begin with the simplest case for V , and work our way to more complicated potentials.

3.3.1 The Free Particle

What's the simplest form of $V(x)$? Well, $V(x) = 0$ of course! This case is known as the *free particle*, which we will study below. In this case, the time-independent Schrodinger equation reads:

$$-\frac{i\hbar}{2m} \frac{\partial^2}{\partial x^2} \psi = E\psi$$

this is a simple second order ODE, whose solutions are complex exponentials. To solve this, let k be defined as:

$$k = \frac{\sqrt{2mE}}{\hbar}$$

so the differential equation now reads:

$$\frac{\partial^2}{\partial x^2} \psi = -k^2 \psi$$

which has a well-known solution of the form:

$$\psi(x) = Ae^{ikx} + Be^{-ikx}$$

which makes sense – these are just plane waves travelling through space unimpeded, which you'd expect if you don't have a potential $V(x)$. In particular, this solution is a superposition of a wave travelling to the right with positive k and one to the left with negative k .

However, this solution has a problem: it violates normalizability! If we try to check the normalization:

$$\int_{-\infty}^{\infty} (Ae^{ikx} + Be^{-ikx}) (A^*e^{-ikx} + B^*e^{ikx}) dx = \int_{-\infty}^{\infty} |A|^2 + |B|^2 + AB^*e^{2ikx} + A^*Be^{-2ikx} dx = \infty$$

which diverges to infinity. So what does this mean? Well, the only natural conclusion is that such a wavefunction is inadmissible, and therefore a free particle with a definite energy E

cannot exist. Instead, the solution is to consider a linear combination of a *range* of energies, represented as an integral:

$$\psi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \phi(k) e^{ikx} dk \quad (3.1)$$

where $\phi(k)$ essentially represents the “weight”, or contribution that the wave with momentum k contributes to the overall $\psi(x)$. Those that are familiar with Fourier analysis will recognize this equation immediately, and point out that $\psi(x)$ and $\phi(k)$ are related by the Fourier transform:

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \psi(x) e^{-ikx} dx$$

And because of this fact, this means that $\psi(x)$ and $\phi(k)$ are actually equivalent descriptions of the same state! This doesn’t resolve our initial problem though: how do we find $\phi(k)$? The trick lies in the fact that $\phi(k)$ has no explicit time-dependence, so we can pick a time which we know the form of $\psi(x)$ (which must be normalized), and solve for $\phi(k)$ there. Generally, this time is chosen to be at $t = 0$, so:

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \psi(x, 0) e^{-ikx} dx$$

which you can then plug into Equation 3.1 to get $\psi(x)$. In essence, because the Fourier transform doesn’t touch the normalization condition at all (it just provides a means to transform from position to frequency space), once you start with a normalized wavefunction, it will always be normalized.

3.3.2 The Infinite Square Well

With $V(x) = 0$ solved, let’s move forward and look at our first non-trivial $V(x)$, defined by the piecewise function:

$$V(x) = \begin{cases} 0 & 0 \leq x \leq a \\ \infty & \text{otherwise} \end{cases}$$

this is called the *infinite-square-well*, appropriately named since the graphical representation of the potential is a well whose walls are infinitely tall:

[INSERT TIKZ HERE]

To solve this differential equation, we will consider $V(x)$ in the interval $[0, a]$ separately from everywhere else, then tie the solutions together. Luckily for us, both solutions are relatively simple. between $[0, a]$, since $V(x) = 0$, then we revert back to the above case of the free particle, where our solution is just travelling plane waves. For reasons that will be apparent later, we will choose the trigonometric representation of a travelling wave:

$$\psi(x) = A \sin(kx) + B \cos(kx)$$

This is an equivalent way to representing a travelling wave, and the conversion is given to us by Euler's formula:

$$e^{ikx} = \cos(kx) + i \sin(kx)$$

Now, outside of $[0, a]$, the potential is infinite, so the only sensible solution for ψ is $\psi = 0$.

Okay, so we have the solutions in these separate regimes, how do we tie them together? The key is to realize that $\psi(x)$ must be *continuous*, since x (the position) is a continuous variable and $|\psi(x)|^2$ must represent the probability density in position. So, since the wavefunction is identically zero outside the well, it must approach zero from inside the well toward the boundaries:

$$\lim_{x \rightarrow 0} \psi(x) = \lim_{x \rightarrow a} \psi(x) = 0$$

First, we handle the $x = 0$ case since that's easy:

$$\psi(0) = A \sin(0) + B \cos(0) \implies B = 0$$

Now for the $x = a$ case:

$$0 = \psi(a) = A \sin(ka)$$

so then ka must be an integer multiple of π , hence $ka = n\pi$ or generally written as $k_n = \frac{n\pi}{a}$, the subscript n -th solution. And here we see a very intriguing property fall out of the mathematics: the simple restriction of our wavefunction in space *forces* our wavefunction to belong to a discrete family of functions, parametrized by the integer n . As a consequence, this also quantizes the allowable energies:

$$E_n = \frac{p^2}{2m} = \frac{\hbar^2 k_n^2}{2m}$$

This is the first result that fundamentally differs from classical notions of thinking, as energy is always thought to be a continuum in classical physics. Finally, all we have to do now is to find the normalization constant A , which we do by enforcing the law of total probability:

$$\int_{-\infty}^{\infty} |A|^2 \sin^2 kx \, dx = |A|^2 \frac{a}{2} \implies |A| = \sqrt{\frac{2}{a}}$$

Therefore, the full solution to the infinite square well is the set of wavefunctions ψ_n :

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x\right)$$

Hopefully through this example you can start to see how the idea of quantization falls out very naturally from the simple restriction that ψ must be continuous: it's an example of how quantum mechanics is fundamentally different from classical mechanics, and yet it makes extremely accurate descriptions of the phenomena around us.

3.3.3 Delta Function Potentials

In this section, we will consider $V(x)$ of a very particular form:

$$V(x) = -a\delta(x)$$

Here, $\delta(x)$ refers to the Dirac delta function, which is quite a funny object. It's not a "function" in the traditional sense, and instead defined as a function whose value is zero everywhere but integrates to 1 over the entire real line. In this sense, it's possible to represent the delta function as:

$$\delta(x) = \begin{cases} \infty & x = 0 \\ 0 & \text{elsewhere} \end{cases}$$

while also satisfying:

$$\int_{-\infty}^{\infty} \delta(x) dx = 1$$

there is no true function that simultaneously satisfies these properties, hence the delta function isn't really considered a function.² However, it often makes an appearance in theoretical and experimental physics since many phenomena around us can be modeled approximately as delta functions.

Just like all of the potentials we've seen so far, let's throw this $V(x)$ into the differential equation:

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} - a\delta(x)\psi = E\psi$$

Now, let's consider two separate cases: when $x < 0$ and $x > 0$. In both these cases, $\delta(x) = 0$, so therefore our differential equation reduces to:

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} = E\psi \implies \frac{\partial^2 \psi}{\partial x^2} = -\frac{2mE}{\hbar^2} \psi$$

Now it's time to mention a detail we haven't really talked about until now: and that is the sign of E that we're dealing with. As you'll notice from the differential equation above, the sign of E determines whether the sign of the prefactor, which impacts the kind of solutions our differential equation yields. In particular, when $E > 0$, then we admit *scattering* solutions, and when $E < 0$ we admit *decaying* solutions.³

²Formally, $\delta(x)$ is called a *distribution* rather than a function, but we won't worry too much about this distinction. It's good enough for our purposes to treat $\delta(x)$ as a function that is zero everywhere except at $x = 0$ where it equals infinity, and that its integral over all space is equal to one.

³In case you're wondering why we haven't dealt with the sign of E until now, it's because we didn't need to. In the free particle example, a particle with negative energy and admits a decaying solution would blow up at either ∞ or $-\infty$, which are not normalizable and thus inadmissible. A negative energy state isn't admissible in the infinite square well potential either, since there's no way for an exponentially decaying function to be identically zero as is required to match the boundary condition on either side of the well.

The decaying solutions are slightly easier to deal with, so let's tackle those first. In this case, $E < 0$ so the prefactor on ψ is positive, hence we can write:

$$\frac{\partial^2 \psi}{\partial x^2} = -\frac{2mE}{\hbar^2} = \kappa^2 \psi$$

where:

$$\kappa = \frac{\sqrt{-2mE}}{\hbar} \quad (3.2)$$

remember, κ is still positive here since E is negative. Just like the free particle solution, both the positive and negative exponential are valid solutions to the differential equation:

$$\psi = Ae^{\kappa x} + Be^{-\kappa x}$$

and here's where our physical boundary conditions come into play. First, let's consider the $x < 0$ case. Here, we require that ψ doesn't blow up as $x \rightarrow -\infty$, which is only possible if the negative exponential vanishes. Likewise, when $x > 0$ we don't want ψ to blow up as $x \rightarrow \infty$, hence the positive exponential term vanishes. Put together, we have the following "pieces" of our wavefunction:

$$\psi_L = Ae^{\kappa x} \quad \psi_R = Be^{-\kappa x}$$

Now, we need to join the two together to get one complete wavefunction. To do this, we enforce our notion of continuity, which is to say that ψ_L and ψ_R must attain the same value at $x = 0$. This is easy; since they both have the same exponential, this is just telling us that $A = B$. So, the wavefunction looks like:

$$\psi = \begin{cases} Ae^{\kappa x} & x < 0 \\ Ae^{-\kappa x} & x > 0 \end{cases}$$

To find A , we enforce normalization. To do this, notice that ψ is symmetric about $x = 0$, so we can write the integral as:

$$1 = \int_{-\infty}^{\infty} |\psi|^2 dx = 2 \int_0^{\infty} |A|^2 e^{-2\kappa x} dx = \frac{|A|^2}{\kappa}$$

So the solution is $|A|^2 = \kappa$, or $A = \sqrt{\kappa}$, and our wavefunction is then:

$$\psi = \sqrt{\kappa} e^{-\kappa|x|}$$

Note that I combined the piecewise expression into a single one with the use of the absolute value on x . But what would be the energy of such a wavefunction? To find it, we have to look more closely at the behavior of ψ around $x = 0$. To find the energy, we need to impose an additional condition that we haven't encountered yet: continuity of ψ' .

First off, we should try and understand the reason why continuity in ψ' is needed, of which there are two. The first comes directly from our time-independent Schrodinger equation. Recall that the equation reads:

$$\left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right) |\psi\rangle = E |\psi\rangle$$

isolating the term on the left, we have:

$$-\frac{i\hbar}{2m} \frac{\partial^2 \psi}{\partial x^2} = (E - V(x))\psi$$

since ψ is continuous then the right side is continuous, which means the left must be also. If ψ'' is continuous, then this implies that ψ' must be differentiable, hence it must be continuous.

But what if $V(x)$ is not continuous? We've seen two examples already where $V(x)$ is not continuous, so what then? As it turns out, even when $V(x)$ is discontinuous, as long as $V(x)$ is not *infinite*, then continuity in $d\psi/dx$ is still guaranteed, the proof of which we outline below:

Theorem 3.2. *For a given potential $V(x)$, as long as $V(x)$ is finite, the wavefunction ψ and its derivative ψ' is always continuous.*

Proof. Consider an arbitrary $V(x)$. Then, Schrodinger's equation reads:

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V(x)\psi(x) = E\psi(x)$$

we can now integrate this equation around an arbitrary point $[a - \epsilon, a + \epsilon]$:

$$-\frac{\hbar^2}{2m} \int_{a-\epsilon}^{a+\epsilon} \frac{\partial^2 \psi}{\partial x^2} dx + \int_{a-\epsilon}^{a+\epsilon} V(x)\psi(x) dx = E \int_{a-\epsilon}^{a+\epsilon} \psi(x) dx$$

The first term is just the derivative $\psi'(a + \epsilon) - \psi'(a - \epsilon)$, and the term on the right hand side vanishes as $\epsilon \rightarrow 0$. So that leaves us with the equation:

$$\psi'(a + \epsilon) - \psi'(a - \epsilon) = \frac{2m}{\hbar^2} \int_{a-\epsilon}^{a+\epsilon} V(x)\psi(x) dx$$

and now we come to the heart of the argument: if $V(x)$ is finite, then the integral on the right vanishes, and we get:

$$\psi'(a + \epsilon) - \psi'(a - \epsilon) = 0$$

as $\epsilon \rightarrow 0$, which is the definition of continuity in ψ' . However, if $V(x)$ is infinite, then this equality fails. Hence, we've proven that ψ' is always continuous, as long as $V(x)$ is not infinite. $\square$

This is also why we didn't have to consider continuity in ψ' when dealing with the infinite square well. Now we return to the delta function. In this case, since $V(x)$ is not finite, a discontinuity in ψ' is allowed, but the discontinuity at $x = 0$ will tell us more about the allowed energy:

$$\psi'(\epsilon) - \psi'(-\epsilon) = -\frac{2m}{\hbar^2} \int_{-\epsilon}^{\epsilon} a\delta(x)\psi(x) dx = -\frac{2ma}{\hbar^2} \psi(0) = -\frac{2ma}{\hbar^2} A \quad (3.3)$$

The derivative of ψ is:

$$\psi' = \begin{cases} A\kappa e^{\kappa x} & x < 0 \\ -A\kappa e^{-\kappa x} & x > 0 \end{cases}$$

and as $x \rightarrow 0$, then $\psi' \rightarrow A\kappa$ for $x < 0$ and $-A\kappa$ for $x > 0$. Substituting these values into Equation 3.3 gives us:

$$-2A\kappa = -\frac{2ma}{\hbar^2}A \implies \kappa = \frac{ma}{\hbar^2} \quad (3.4)$$

which means we can finally solve for E ! Recall all the way back (Equation 3.2) that we defined E in terms of κ . But now since we know κ in terms of other constants, we can finally back out E :

$$E = -\frac{ma^2}{2\hbar^2}$$

and so finally, we have the full solution for the delta function potential, for $E < 0$:

$$\psi(x) = \frac{\sqrt{ma}}{\hbar} e^{-ma|x|/\hbar^2}, \quad E = -\frac{ma^2}{2\hbar^2}$$

Now, we have to deal with the scattering states $E > 0$. In this case, the solutions at $x < 0$ and $x > 0$ both admit plane wave solutions, one traveling to the left and another to the right:

$$\psi(x) = Ae^{-ikx} + Be^{ikx}, \quad k = \frac{2mE}{\hbar}$$

The coefficients at this moment are undetermined, so we have four constants to deal with:

$$\psi_L(x) = Ae^{-ikx} + Be^{ikx} \quad \psi_R(x) = Ce^{-ikx} + De^{ikx}$$

We now know how to combine them into one big equation: we first enforce continuity at $x = 0$, which gives us the equation $A + B = C + D$. Now, we enforce continuity in the derivatives. They are:

$$\begin{aligned} \psi'_L(0) &= ik(B - A) \\ \psi'_R(0) &= ik(D - C) \end{aligned}$$

Employing the same trick as Equation 3.3, we get:

$$\psi'(\epsilon) - \psi'(-\epsilon) = ik(D - C) - ik(B - A) = \frac{2ma}{\hbar^2}\psi(0) = \frac{2ma}{\hbar^2}(A + B)$$

We choose to use $\psi_L(0) = A + B$ for the last equality; the logic works the same if you use ψ_R . There is a more compact way to write this equality, if you let $\beta = \frac{ma}{\hbar^2 k}$ then you have:

$$C - D = A(2i\beta + 1) + B(2i\beta - 1)$$

So we have the following two equations for A, B, C and D :

$$\begin{aligned} A + B &= C + D \\ D - C &= B(1 - 2i\beta) - A(1 + 2i\beta) \end{aligned}$$

But this is a big problem! We have 4 constants but only two equations, so we'll never be able to fully solve for all constants. There is a solution though – and the key is to appeal to our intuition about what system we're trying to describe. Say we're sending in a wave from the left. Then, it would not make sense to consider a left-traveling wave in ψ_R , so that means that $C = 0$, and from here we solve for A and D in terms of B . If you work out the algebra, you get the following:

$$A = -B \frac{i\beta}{1 + i\beta}, \quad D = \frac{B}{1 + i\beta}$$

So what do we make of these solutions? By solving for the coefficients A and D , they tell you the amplitude of the reflected and transmitted wave when you send in a wave with amplitude B from the left. In particular, we can now calculate quantities such as the reflection coefficient, which gives you the probability that a wave is reflected:

$$R = \frac{|A|^2}{|B|^2} = \frac{\left| -B \frac{i\beta}{1 + i\beta} \right|^2}{|B|^2} = \frac{\beta^2}{1 + \beta^2}$$

and similarly, we can calculate a transmission coefficient:

$$T = \frac{|D|^2}{|B|^2} = \frac{\left| \frac{B}{1 + i\beta} \right|^2}{|B|^2} = \frac{1}{1 + \beta^2}$$

what's nice is that we can verify that the reflection and transmission add up to 1:

$$R + T = \frac{\beta^2}{1 + \beta^2} + \frac{1}{1 + \beta^2} = 1$$

which makes sense – the wave is either transmitted or reflected, without loss. If you now bring back the substitution for β that we did earlier, we can write these in terms of the energy itself:

$$R = \frac{1}{1 + (2\hbar^2 E / ma^2)}, \quad T = \frac{1}{1 + (ma^2 / 2\hbar^2 E)}$$

and that's the full solution to the delta barrier! Firstly, you can already see that this derivation is *very* non-classical: it predicts that a potential well has the chance to scatter a particle

At this point, let's talk briefly about the delta potential *barrier*, which is basically the same potential but positive: $V(x) = a\delta(x)$. While it may seem like we need to do the math all over again, we actually don't since switching the sign only changes the behavior at $x = 0$, so our scattering state remains the same. What does change though is the bound state. To see why, recall that for the bound state, κ was defined as (Equation 3.2):

$$\kappa = \frac{\sqrt{-2mE}}{\hbar}$$

which was guaranteed to be *real* and *positive*, since $E < 0$. However, if we look at our other definition of κ in terms of a and m (Equation 3.4):

$$\kappa = \frac{ma}{\hbar^2}$$

and so if $a < 0$, this solution for κ becomes inconsistent with the first, and hence such a bound state cannot exist. However, the scattering solutions are fine because we don't have an equivalent restriction on k .

Remark. It is interesting to note that because the transmission and reflection are dependent on a^2 , these quantities remain the same regardless of a well or a barrier! This is quite striking – for one, just the mere fact that two different potentials give you the same reflection and transmission is fascinating. Moreover, the fact that a potential well can induce a reflection is also fascinating. Classically, this would be the equivalent of throwing a ball over a well and it having a chance to bounce right back at you, but in quantum mechanics, such a phenomenon is totally allowed!

3.3.4 The Potential Step

Here, we will consider a potential represented by a wall of finite height:

$$V(x) = \begin{cases} 0 & x < 0 \\ V_0 & x > 0 \end{cases}$$

There are two different kind of states to consider, $0 < E < V_0$ and $E > V_0$, and the $E < 0$ state does not exist. Let's consider the $E < V_0$ state first. To the left of the barrier, we have a scattering state, but to the right we have a *decaying* state, since $E - V_0$ is negative. This is because inside the barrier, Schrodinger's equation reads:

$$\frac{\partial^2 \psi}{\partial x^2} = (E - V_0)\psi$$

so following the same substitution for k as in Section 3.3.1, you get:

$$k = \frac{\sqrt{2m(E - V_0)}}{\hbar} = \frac{i\sqrt{2m(V_0 - E)}}{\hbar}$$

since k is purely imaginary, this turns the oscillatory e^{ikx} into a decaying exponential. Combining this with the oscillatory solution to the left of the barrier gives you the following solution for ψ :

$$\psi(x) = \begin{cases} Ae^{ikx} + Be^{-ikx} & x < 0 \\ Ce^{\kappa x} + De^{-\kappa x} & x > 0 \end{cases}$$

with k and κ being defined as:

$$k = \frac{\sqrt{2mE}}{\hbar} \quad \kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar}$$

Again, we're faced with a situation where continuity alone doesn't give us enough equations to fully solve, and have to be more specific about the problem to eliminate some variables. Firstly, let's say that our wave is coming in from the left. This would mean that inside the barrier, an exponentially *growing* solution would not make sense, and therefore we can set $C = 0$. Now, we have enough equations to solve for B and D in terms of A . In particular, at $x = 0$, the continuity constraints in ψ and ψ' gives us the following system of equations:

$$\begin{aligned} A + B &= D \\ ik(A - B) &= -\kappa D \end{aligned}$$

Now, expressing B and D in terms of A (remember, A is our incident wave amplitude, so it makes sense to solve in terms of A) just requires some algebra, and ultimately gets you the relations:

$$B = \frac{ik + \kappa}{ik - \kappa} A, \quad D = \frac{2ik}{ik - \kappa} A$$

Here, we can calculate the reflection coefficient:

$$R = \frac{|B|^2}{|A|^2} = \left| \frac{ik + \kappa}{ik - \kappa} \right|^2 = 1$$

so this tells us that the entire amplitude of the wave is reflected, which makes sense since the barrier is infinitely thick, so any wave will be fully reflected back. As we'll see in Section 3.3.5, when the barrier is only finitely thick you will get a transmitted wave, called *tunnelling*.

You may be wondering: if the entire wave is reflected, then why is the wavefunction nonzero inside the barrier? Doesn't this imply that there is a nonzero probability for the particle to *be* inside the barrier, and thus not be reflected?

The answer to this question is quite subtle. Firstly, decaying solutions don't carry any energy, as their probability current turns out to be zero. Hence, all the energy must be reflected back. But, perhaps this isn't a satisfying answer, in which case there another solution. Suppose you *were* able to measure the particle inside the barrier. Well, the decaying solution has a characteristic length of $\frac{1}{\kappa}$, so your variance in position is $\Delta x \sim \frac{1}{\kappa}$. But, by the uncertainty relation, you find that:

$$\Delta p_x \geq \frac{\hbar}{\Delta x} = \hbar \kappa = \sqrt{2m(V_0 - E)}$$

and plugging this into the energy:

$$\Delta E = \frac{(\Delta p_x)^2}{2m} \geq V_0 - E$$

and this tells us that by measuring your particle inside the barrier, you've inadvertently *scattered* your particle, and it's no longer in the barrier! So, it's physically impossible to measure your particle to be inside the barrier, no matter how careful you are.

Now we consider the second solution, the $E > V_0$ case. In this case, both $x < 0$ and $x > 0$ admit scattering solutions, but with different wavenumbers:

$$\psi(x) = \begin{cases} Ae^{ikx} + Be^{-ikx} & x < 0 \\ Ce^{ikx} + De^{-ikx} & x > 0 \end{cases}, \quad k = \frac{\sqrt{2mE}}{\hbar}, \quad k' = \frac{\sqrt{2m(E - V_0)}}{\hbar}$$

again, we impose a physical constraint, of considering a particle incident from the left. This kills the left travelling wave in $x > 0$, so $D = 0$. Using the continuity constraints again, we get the following relation:

$$B = \frac{kk'}{k + k'}A \quad C = \frac{2k}{k + k'}A$$

This algebra follows very similarly from the previous examples, so we'll skip showing it here. With this, we can now calculate the reflection and transmission:

$$R = \frac{|B|^2}{|A|^2} = \frac{\left[1 - \left(1 - \frac{V_0}{E}\right)^{1/2}\right]^2}{\left[1 + \left(1 - \frac{V_0}{E}\right)^{1/2}\right]^2} \quad T = \frac{4\left(1 - \frac{V_0}{E}\right)^{1/2}}{\left[1 + \left(1 - \frac{V_0}{E}\right)^{1/2}\right]^2}$$

and you should verify on your own that $R + T = 1$. Just like the delta potential well, we find that even though the energy of your particle can be higher than the step, you *still* get a reflection regardless, which is a fundamentally non-classical result.

3.3.5 The Potential Barrier

This potential is similar to that of the potential step, but this time it's a finitely long wall:

$$V(x) = \begin{cases} 0 & x < 0 \\ V_0 & 0 < x < a \\ 0 & x > a \end{cases}$$

visually, it looks like this:

[INSERT TIKZ HERE]

Here, there are two cases we need to consider: $E > V_0$ and $0 < E < V_0$. For $E < 0$, we get an infinitely decaying solution, which is to say that $\psi = 0$ in that case. Let's start with the $0 < E < V_0$ case, which as we mentioned before, will lead to a tunnelling situation. We'll split up the wavefunctions into three regions:

1. To the left of the wall: $x < 0$

2. Above the wall: $0 < x < a$
3. To the right of the wall: $x > a$

So the respective wavefunctions we solve for will be labeled ψ_1 , ψ_2 and ψ_3 according to the above numbering. Starting off with $x < 0$, we have the same plane wave solution as before:

$$\psi_1(x) = Ae^{ikx} + Be^{-ikx}$$

with $k = \frac{\sqrt{2mE}}{\hbar}$ as before. The same goes for the right:

$$\psi_3(x) = Ce^{ikx} + De^{-ikx}$$

Now let's look at the Schrodinger equation for $0 < x < a$:

$$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi + V_0 \psi = E\psi$$

Moving the V_0 to the right gives:

$$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi = (E - V_0)\psi$$

And since $E < V_0$ here, then we have exponentially decaying solutions inside the wall:

$$\psi_2(x) = Fe^{\kappa x} + Ge^{-\kappa x}$$

with $\kappa = \sqrt{2m(V_0 - E)}/\hbar$. Unlike the potential step though, we can't simply eliminate the exponentially growing solution by setting $F = 0$. Because the barrier isn't infinitely long, the exponentially growing solution is allowed. However, we can still eliminate D , by considering a particle incident from the left.

Now, because we have split ψ into three wavefunctions, we need to stitch them together by enforcing continuity in ψ and ψ' at $x = 0$ and $x = a$. The $x = 0$ conditions are:

$$\begin{aligned} A + B &= F + G \\ ik(A - B) &= \kappa(F - G) \end{aligned}$$

and similarly, the $x = a$ ones are:

$$\begin{aligned} Ce^{ika} &= Fe^{\kappa a} + Ge^{-\kappa a} \\ ike^{ika} &= \kappa(Fe^{\kappa a} - Ge^{-\kappa a}) \end{aligned}$$

From here, we can then solve for $\frac{B}{A}$ and $\frac{C}{A}$, which will give us the reflection and transmission coefficients. At this point, the algebra also gets quite long so we won't go into too much detail, but there is a relatively "nice" solution involving hyperbolic sine and cosine:

$$R = \left[1 + \frac{4E(V_0 - E)}{V_0^2 \sinh(ka)} \right], \quad T = \left[1 + \frac{V_0^2 \sinh^2(\kappa a)}{4E(V_0 - E)} \right]$$

Now, we turn our attention to the $E > V_0$ case. Here, all three regions have scattering solutions, but k is different above the barrier because of V_0 :

$$\psi = \begin{cases} Ae^{ikx} + Be^{-ikx} & x < 0 \\ Fe^{ik'x} + Ge^{-ik'x} & 0 < x < a \\ Ce^{ikx} + De^{-ikx} & x > a \end{cases}$$

Again, follow the same algebra and you'll get:

$$R = \left[1 + \frac{4E(E - V_0)}{V_0^2 \sin^2(k'a)} \right]^{-1} \quad T = \left[1 + \frac{V_0^2 \sin^2(k'a)}{4E(E - V_0)} \right]^{-1}$$

Here, we want you to focus on the physics and not the algebra so we won't go into detail, but it really does work in the same way we've always been doing it: consider a left-incident wave, set $D = 0$, then eliminate F and G and solve for $\frac{B}{A}$ and $\frac{C}{A}$. Finally, square these and you get your reflection and transmission coefficient.

Speaking of focusing on the physics, notice that because the transmission term has a $\sin^2(k'a)$ term, it means that there are some configurations where the $T = 1$ and $R = 0$, hence you get perfect transmission!

3.3.6 Finite Square Well

Finally, the last potential we will look at in this chapter is the finite square well. This is similar to the infinite square well, but this time our walls are finitely tall. As we'll see, this gives us a first glimpse into why the Schrodinger equation is difficult to solve analytically. Mathematically, we will express this potential as:

$$V(x) = \begin{cases} -V_0 & -a < x < a \\ 0 & \text{otherwise} \end{cases}$$

and here's a visual picture of what the well looks like:

[INSERT TIKZ HERE]

There are two cases we have to consider: the $-V_0 < E < 0$ bound state, and $E > 0$, the scattering state.

Starting with the first case, let's again break up ψ into two parts: inside and outside the well. Inside the well, the Schrodinger equation is written as:

$$\frac{\partial^2 \psi}{\partial x^2} + \alpha^2 \psi = 0$$

with α defined as:

$$\alpha = \left[-\frac{2m}{\hbar^2} (V_0 + E) \right]^{1/2}$$

This is a scattering solution, since the energy of the particle is larger than $-V_0$, so it will give us oscillatory solutions. For reasons that will become clear in a moment, we will write ψ in this region in terms of sines and cosines:

$$\psi = A \cos(\alpha x) + B \sin(\alpha x), |x| < a$$

Outside the well, we have decaying solutions since $E < 0$. The Schrodinger equation in this region reads as:

$$\frac{\partial^2 \psi}{\partial x^2} - \beta \psi(x) = 0$$

and β is:

$$\beta = \left(-\frac{2mE}{\hbar^2} \right)^{1/2}$$

Now, we will leverage the following theorem to make our lives easier:

Theorem 3.3 (Wavefunction for Symmetric Potentials). *If the potential $V(x)$ satisfies $V(x) = V(-x)$, then its wavefunction also satisfies $\psi(x) = \psi(-x)$.*

Proof. Suppose first that $\psi(x)$ satisfies the Schrodinger equation for $V(x)$. That is, $\psi(x)$ is a solution to:

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi(x)}{\partial x^2} + V(x)\psi(x) = E\psi(x) \quad (3.5)$$

Now, consider $\psi(-x)$:

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi(-x)}{\partial x^2} + V(-x)\psi(-x) = E\psi(-x)$$

since $V(-x) = V(x)$, then:

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi(-x)}{\partial x^2} + V(x)\psi(-x) = E\psi(-x)$$

and since this differential equation is the same as the original equation (Equation 3.5), then $\psi(-x)$ is also a solution. Since this is the case, then any even and odd linear combination of these will also be a solution. That is, the functions:

$$\begin{aligned} \phi(x) &= \psi(x) + \psi(-x) \\ \phi'(x) &= \psi(x) - \psi(-x) \end{aligned}$$

are also solutions. Therefore, we can always take ψ to be even or odd. □

In our case, the finite square well indeed does satisfy this property, and so we can focus on looking for even and odd solutions. This makes our lives much easier, since it means that we only need to consider one set of boundary conditions at $x = a$, instead of two.

Let's start by considering ψ even. Then, the wavefunction is:

$$\psi(x) = \begin{cases} B \cos(\alpha x) & 0 < x < a \\ C e^{-\beta x} & |x| > a \end{cases}$$

now, imposing boundary conditions at $x = a$ gives:

$$\begin{aligned} \psi : A \cos(\alpha a) &= C e^{-\beta a} \\ \psi' : -\alpha A \sin(\alpha a) &= -\beta C e^{-\beta a} \end{aligned}$$

dividing one equation by the other gives:

$$\alpha \tan(\alpha a) = \beta$$

similarly, for the odd solutions, we kill the $B \cos(\alpha x)$ term in $0 < x < a$ and the decaying solution remains the same. Applying the same boundary conditions gives you the equation:

$$\alpha \cot(\alpha a) = -\beta$$

both of these equations represent the condition for the allowed energies of the system. That is, for a given a and V_0 , these equations tell you which energies E are allowed. Once you figure that out, you can then normalize by enforcing that it integrates to 1. What's interesting about these equations is that they're not analytically solvable, so you can only solve this numerically.

Remark. Hopefully, by working out these bound states you can get a better appreciation for why the Schrodinger equation is typically a very difficult equation to solve analytically. The potential we're dealing with here is hardly anything complicated – it's just a simple square well of finite depth, yet we're already having to resort to numerical methods to get the allowed energies.

As you can guess, this becomes a general trend as we move toward more complicated potentials. Rarely are any of these potentials analytically solvable, so we have to resort to numerical methods or find a way to approximate the solutions. This latter idea is the central idea of the second part of this book, [INSERT PART 2 REFERENCE].

Now let's deal with the scattering solution. This ends up working out similarly to the delta potential barrier and well, in that the math actually works out the same way. For the left incident particle, we have:

$$\psi(x) = \begin{cases} A e^{ikx} + B e^{-ikx} & x < -a \\ F e^{ik'x} + G e^{-ik'x} & -a < x < a \\ C e^{ikx} & x > a \end{cases}$$

there is no left travelling wave at $x > a$ since we are considering a left-incident wave. Notice that this is exactly the same situation as the finite potential barrier, except k' is *slightly* different,

in that we make the swap $V_0 \rightarrow -V_0$. All the algebra remains the same, so we can just copy over the reflection and transmission coefficient we got from there and make this simple edit to V_0 .⁴

$$R = \left[1 + \frac{4E(E + V_0)}{V_0^2 \sin^2(2k'a)} \right]^{-1} \quad T = \left[1 + \frac{V_0^2 \sin^2(2k'a)}{4E(V_0 + E)} \right]^{-1}$$

3.4 Time-Dependent Schrodinger Equation

So far, we've been dealing with the time *independent* Schrodinger equation, which as its name suggests, does not include time as a parameter. In this section, we will look at how we transform a solution to the time-independent solution into one that satisfies time dependence. While this may seem complicated at first, the trick actually turns out to be very simple.

3.4.1 Separable solutions

To begin, let's take a look at the time-dependent Schrodinger equation:

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V\Psi$$

Because we deal with time-independent potentials, it means that we can actually solve this equation by method of *separation of variables*. That is, we look for solutions of the form:

$$\Psi(x, t) = \psi(x)\phi(t)$$

solving for $\psi(x)$ was pretty much all that we've been doing in this chapter so far. Now, given a $\psi(x)$, what is the appropriate $\phi(t)$ to tack on so that $\Psi(x, t)$ is a solution to the time-dependent equation? Let's first plug this separable solution back into the Schrodinger equation:

$$i\hbar \psi \frac{d\phi}{dt} = -\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} \phi + V\psi\phi$$

dividing through by $\psi\phi$:

$$i\hbar \frac{1}{\phi} \frac{d\phi}{dt} = -\frac{\hbar^2}{2m} \frac{1}{\psi} \frac{d^2\psi}{dx^2} + V \quad (3.6)$$

now comes a crucial argument: the left hand side is a function of t , while the right side is a function of x . Therefore, in order for this equation to hold both sides **must** be constant. Otherwise, you could change one side of this equation without changing the other, violating the equality. Therefore, we can set both sides to be constant:

$$i\hbar \frac{1}{\phi} \frac{d\phi}{dt} = E \implies \frac{d\phi}{dt} = -\frac{iE}{\hbar} \phi$$

⁴we also have to make the change $a \rightarrow 2a$ compared to the square barrier since a was the width of the barrier, but a as defined in the finite well is only half the width of the well.

this is now a simple differential equation for ϕ , which gives solutions:

$$\phi(t) = e^{-iEt/\hbar}$$

and that's our time dependence! All you have to do now is throw this onto every solution to the spatial $\psi(x)$, and you've got your general solution $\Psi(x, t)$. As a side note, setting the right hand side of Equation 3.6 to E gives you the time-independent Schrodinger equation!

At this point, you may also be wondering if we've covered *all* possible solutions this way. After all, the condition that $\Psi(x, t)$ be separable is a fairly restrictive one. As it turns out, whenever the Hamiltonian is time-independent and $\hat{H}$ admits a complete set of energy eigenfunctions, then the separable solutions are enough to describe any general solution. Here, completeness refers to the condition that any square-integrable function (i.e. any function in our Hilbert space $\mathcal{H}$) can be written as a linear combination of these eigenfunctions. Since every solution $\Psi(x, t)$ must be square-integrable, then this completeness condition is all we need to guarantee that we've covered all solutions.

Remark. This completeness condition is actually not as restrictive as you might think – all the Hamiltonians we will deal with in this book satisfy this completeness, even though we'll be looking at some pretty complicated potentials later on!

References